# Event analysis: selection and pad histories

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The integrated checkpoint reads source-frame hits, produces rounded ion counts, samples a common drift distance, diffuses each ion, and constructs Petit's kNN graph. Events pass when exactly one component survives the configured ion threshold. Accepted events supply their full retained diffusion table to pad counting. The cumulative Julia suite passes 2016 assertions.

## Selection and exact charge accounting

| Sample | Accepted/tested | Retained ions | Pad ions | Below cut, saved | Mean ms |
|---|---|---|---|---|---|
| Double beta | 22/50 | 2468695 | 2468695 | 7393 | 54.8 |
| Electron background | 31/50 | 3474395 | 3474395 | 14821 | 49.3 |

These events are a reproducible validation subset spread across each input file; the acceptance fractions describe this subset. Retained and below-cut totals in the table refer to accepted events. Every accepted event satisfies $N_{\text{pads}} = N_{\text{diffusion,retained}}$ exactly. Rejected events retain their decision and diffusion ledger. All events satisfy $N_{\text{sampled}} = N_{\text{retained}} + N_{\text{outside}}$ and $N_{\text{retained}} = N_{\text{graph}} + N_{\text{below threshold}}$. Timings cover the computational pipeline after a warm-up, including allocations; file reads and writes are outside the timed region.

## Model and conventions

The model retains the common-event thermal diffusion and per-hit rounding from step two. Reconstruction cells have width $3\sigma_T$, maximum graph edges $6\sigma_T$, and the configured nearest-neighbour count. The graph threshold is applied to summed Int64 ion counts, inclusively; energies retain Petit's Float32 weights for the backbone. This gives exact equality at the configured threshold.

The selected track's energy-weighted x/y centre and minimum z define the grid references, following Petit. Transverse grid edges receive one independently sampled offset per axis in [0,pitch), shared by all slices. Whole-cell extensions cover every retained diffusion-bin centre. All ions at a diffusion-bin centre are assigned to its containing cell: this is the geometric collection approximation. The independent physical pitch is read from the detector card. Internal cells are half open; the final upper edge is included. Sparse pad histories retain zero-count intervals through their complete saved axes.

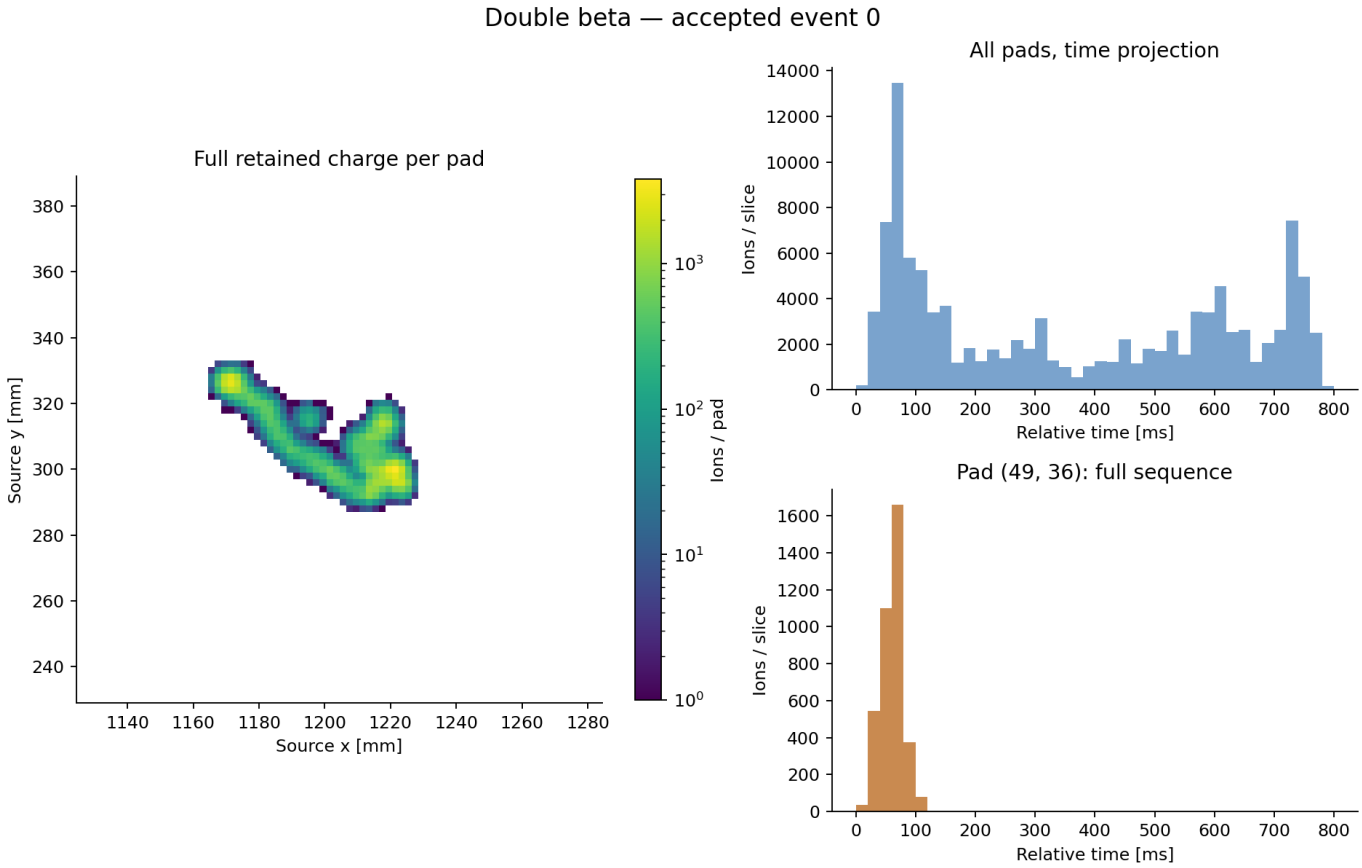
The relative-time direction is increasing source z gives later arrival. The first arrival edge is t=0; the sign and its source-z coordinate are saved with each history. Both signs are supported. With 2 mm slices and 0.1 mm/ms drift, each slice is 20 ms. Absolute event arrival offsets are a separate convention from these saved relative times. Drift, diffusion and grid RNG streams use SHA-256 keys built from run seed, sample, event ID and stage; event processing order preserves each stream.

## Validation

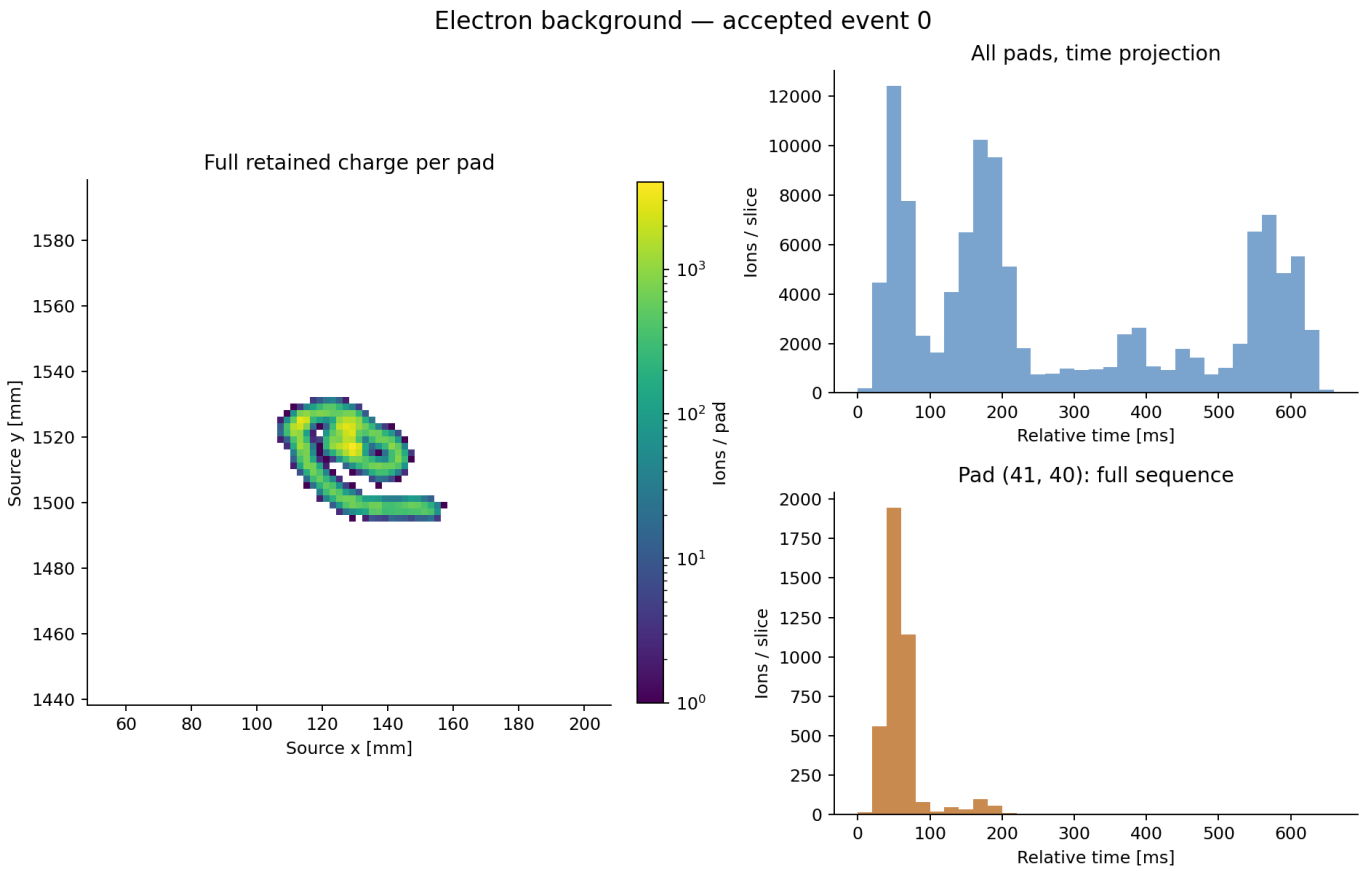
Tests cover Petit occupancy parity away from its excluded final boundaries, exact threshold equality, disconnected-event rejection, full retained-charge collection, grid extension, integer-pitch phase shifts, final edges, empty histories, zero slices, both time directions and input ownership. An end-to-end synthetic

signal passes and a two-cloud background fails. Real-event replay is identical; saved HDF5 histories are read back to check charge conservation.

### Double beta; event 0



## Electron background; event 0



## Reproduction and next increment

```
make validate-event-step3 PYTHON=/path/to/python3.11-or-newer
# Presentation only:
make report-event-step3 PYTHON=/path/to/python3.11-or-newer
```

Numerical tables, histories, diffusion tables, axes, seeds and source hashes are in:
`results/event_analysis/step3_validation.h5`.

The next increment establishes the production output schema and common ensemble axes/denominators, then runs parameter scans and the full samples.

## Configuration used

```
# Event-analysis checkpoints. Detector constants come from the SI card.
[inputs]
detector_card = "config/detector.toml"
samples = ["events/0nubb_hitdf.h5", "events/electron_0nubb_energy_hitdf.h5"]

[ionization]
ionisation_energy_eV = 21.9
# Same adopted nominal W as the initial ion_tracks analysis.

[drift]
min_cm = 10.0
max_cm = 150.0

[diffusion]
longitudinal = "thermal" # "zero" is the longitudinal comparison
nbins = 100
nsigma = 3.0

[graph]
method = "kNN"
k = 10
min_ions_per_bin = 10
bin_sigma_scale = 3.0
edge_sigma_scale = 6.0

[pads]
z_bin_width_mm = 2.0
minimum_xy_span_mm = 160.0
random_grid_offset = true
# Relative-time convention for this checkpoint: increasing z is later.
# Change to -1 for decreasing-z arrival order. t=0 is the first arrival edge.
drift_direction = 1
# Transverse pitch and ion speed are read from the detector card.

[random]
seed = 2026
# Validation uses explicit per-case seeds, stored in its HDF5.

[validation]
events_per_sample = 5
selection_events_per_sample = 50
synthetic_ions = 200000
drift_cm = [10.0, 80.0, 150.0]
bin_counts = [50, 100, 200]
padding_sigma = [2.0, 3.0, 4.0, 5.0]
```